Huawei Access Node QoS Congestion Event Resolution SOP (YEL-ACC-67)

Node: YEL-ACC-67 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_8971

Description

This Standard Operating Procedure (SOP) is designed to troubleshoot and resolve Quality of Service (QoS) congestion events on Huawei Access Nodes, specifically the YEL-ACC-67 node located at the Yelahanka Sub-station. The failure domain in question is attributed to microbursts, interface tail drops, and QoS queue starvation, which are all indicative of a congested network condition. The root cause of these issues can be attributed to the Huawei Access Node's reliance on the QoS scheduling algorithm, which prioritizes traffic based on the DiffServ (Differentiated Services) Code Point (DSCP) values assigned to each packet. When the node experiences a surge in traffic, the QoS queues can become congested, leading to microbursts and interface tail drops. This congestion event can be exacerbated by factors such as high bandwidth utilization, bursty traffic, and misconfigured QoS policies. To resolve this issue, the Huawei Access Node must be reconfigured to optimize QoS performance, which involves adjusting the QoS scheduling algorithm, re-queuing traffic, and reconfiguring QoS policies to prevent future congestion events.

Pre-requisite Checks

- Verify the site location and node ID match the reported failure context (YEL-ACC-67 at Yelahanka Sub-station)
- Check the redundant link status to ensure it is operational and not contributing to the congestion event
- Verify the QoS policies are properly configured and not causing the congestion event

Potential Root Causes

Cause	Probability
Misconfigured QoS policies causing QoS queue starvation	High
High bandwidth utilization leading to interface tail drops	Medium
Bursty traffic causing microbursts	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy, Expected Output: QoS policy configuration information, Description: This command displays the current QoS policy configuration, which can help identify misconfigured policies contributing to the congestion event.

Command: display interface, Expected Output: Interface status and statistics, Description: This command displays the interface status and statistics, which can help identify high bandwidth utilization or interface tail drops.
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```
Command: display queue, Expected Output: Queue status and statistics, Description: This command displays the queue status and statistics, which can help identify QoS queue starvation or misconfigured QoS policies.
```

Resolution Steps

Step 1: Step: Optimize QoS performance by adjusting the QoS scheduling algorithm, Command: qos policy 1 scheduling algorithm diffserv, Description: This command adjusts the QoS scheduling algorithm to prioritize traffic based on DSCP values, which can help prevent QoS queue starvation and congestion events.

Step 2: Step: Re-queue traffic to prevent QoS queue starvation, Command: queue re-queue, Description: This command re-queues traffic to prevent QoS queue starvation and congestion events.

Step 3: Step: Reconfigure QoS policies to prevent future congestion events, Command: qos policy 1, Description: This command reconfigures QoS policies to prevent future congestion events by adjusting the QoS scheduling algorithm, re-queuing traffic, and optimizing QoS performance.

Post-Check Verification

```
Command: display qos policy, Expected Output: Optimized QoS policy configuration information, Description: This command verifies that the QoS policy configuration has been optimized to prevent QoS queue starvation and congestion events.

Command: display interface, Expected Output: Optimized interface status and statistics, Description: This command verifies that the interface status and statistics have been optimized to prevent high bandwidth utilization and interface tail drops.

Command: display queue, Expected Output: Optimized queue status and statistics, Description: This command verifies that the queue status and statistics have been optimized to prevent QoS queue starvation and misconfigured QoS policies.
```

Roll-back Procedures

- Step: Restore previous QoS policy configuration, Command: qos policy 1 restore, Description: This command restores the previous QoS policy configuration to prevent any negative impact on the network.

- Step: Revert QoS scheduling algorithm changes, Command: qos policy 1 scheduling algorithm revert, Description: This command reverts the QoS scheduling algorithm changes to prevent any negative impact on the network.

- Step: Revert re-queue changes, Command: queue re-queue revert, Description: This command reverts the re-queue changes to prevent any negative impact on the network.